

7.1 Fig. 1 shows an amplifier, in which M1 and M2 have equal size of $50\mu\text{m}/0.5\mu\text{m}$ and the load resistor $R_D=2\text{ k}\Omega$. The amplifier is biased with $I_D=1\text{ mA}$ and $V_{DD}=3\text{V}$, and all transistors are in saturation region. Neglect all the parasitic capacitors. The device parameters are $\mu_n C_{OX}=1.34\times 10^{-4}\text{ A/V}^2$, $V_{TH}=0.7\text{V}$, and $\lambda=0$, the noise coefficient $\gamma=2/3$. Calculate the total input-referred **thermal noise voltage** in a 100-MHz bandwidth? (HINT: You should integrate the obtained input referred noise power density in the bandwidth of 0~100MHz and the noise voltage is the square root the integration results.)

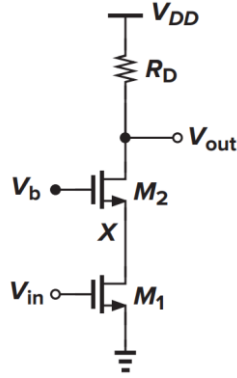


Fig. 1

$$\bar{V}_{n,in}^2 = 4kT \left(\frac{\gamma}{g_{m1}} + \frac{1}{g_{m1}^2 R_D} \right)$$

详细见PPT49页，其中M2的噪声可以忽略

M2 噪声可以忽略的原因是把 M2 的栅极噪声传递到 V_{out} 时，需要把 M2 管子看成输入端，则这个放大器的等效跨导非常小，为 $\frac{g_{m2}}{1 + g_{m2}r_{o1}}$ ，则 M2 的栅极噪声传递到输出端为： $\frac{4kT\gamma}{g_{m2}} \frac{g_{m2}^2 R_D^2}{(1 + g_{m2}r_{o1})^2} \approx \frac{4kT\gamma R_D^2}{g_{m2}r_{o1}^2}$ ，这个值是非常小的，所以可以忽略

$$g_{m1} = \sqrt{2I_D \mu_n C_{OX} \left(\frac{W}{L} \right)_1} = 5.18\text{mS}$$

$$k = 1.38 \times 10^{-23}\text{ J/K} \quad T = 300\text{K}$$

$$\bar{V}_{n,in}^2 = 2.44 \times 10^{-18}\text{ V}^2/\text{Hz}$$

$$V_{in,noise} = \sqrt{\int_0^{100\text{MHz}} \bar{V}_{n,in}^2 df} = 1.562 \times 10^{-5}\text{ V}$$

7.2 Consider the common-source stage of Fig. 7.2 Assume that $(W/L)_1 = 50/0.5$, $I_{D1} = I_{D2} = 0.1$ mA, and $V_{DD} = 3$ V. The device parameters are $\mu_n C_{OX} = 1.34 \times 10^{-4} \text{ A/V}^2$, $\mu_p C_{OX} = 0.38 \times 10^{-4} \text{ A/V}^2$, $V_{THN} = 0.7 \text{ V}$, $V_{THP} = -0.8 \text{ V}$, the noise coefficient $\gamma = 2/3$. If the contribution of M2 to the **input-referred noise voltage (not voltage squared)** must be **one-third** of that of M1, please calculate the maximum output voltage swing of the amplifier.

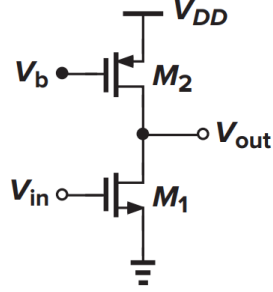


Fig. 7.2

根据 M2,M1 输入噪声比值 1:3 我们可以得到: (输入噪声的比例和转化到输出噪声的比例是一样的)

$$4kT\gamma g_{m2}(r_{o1} \parallel r_{o2})^2 = \frac{1}{9} \times 4kT\gamma g_{m1}(r_{o1} \parallel r_{o2})^2 \Rightarrow g_{m2} = \frac{1}{9} g_{m1}$$

$$g_{m1} = \sqrt{2I_D \mu_n C_{OX} \left(\frac{W}{L}\right)_1} = 1.64 \text{ mS}$$

$$V_{out,max} = V_{DD} - V_{OD2} = V_{DD} - \frac{2I_D}{g_{m2}} = V_{DD} - \frac{18I_D}{g_{m1}}$$

$$V_{out,min} = V_{OD1} = \frac{2I_D}{g_{m1}}$$

$$Swing = V_{out,max} - V_{out,min} = V_{DD} - \frac{20I_D}{g_{m1}} = 1.78 \text{ V}$$

7.3 For the source degenerated common-source stage and folded cascode amplifier shown in Fig. 7.3, derive the expression for input-referred **thermal noise power density** if $\lambda = \gamma = 0$.



Fig. 7.3

(a)

$$\overline{V_{n,out}^2} = 4kTR_D + 4kTR_S A_V^2 + \frac{4kT\gamma}{g_{m1}} A_V^2$$

$$A_V = \frac{g_{m1} R_D}{1 + g_{m1} R_S}$$

$$\overline{V_{n,in}^2} = \frac{4kT(1 + g_{m1} R_S)^2}{g_{m1}^2 R_D} + 4kTR_S + \frac{4kT\gamma}{g_{m1}}$$

(b)

M₂ 的噪声可以忽略，我们可以先把 M₁ 的噪声等效到输出端

$$\overline{V_{n,in,M1}^2} = \frac{4kT\gamma}{g_{m1}} \Rightarrow \overline{V_{n,out,M1}^2} = \frac{4kT\gamma}{g_{m1}} \times g_{m1}^2 R_D^2 = 4kT\gamma g_{m1} R_D^2$$

$$\overline{V_{n,out,R_D}^2} = 4kTR_D$$

$$A_V = -g_{m3} R_D$$

$$\overline{V_{n,in}^2} = \frac{4kT\gamma}{g_{m3}} + \frac{\overline{V_{n,out,R_D}^2} + \overline{V_{n,out,M1}^2}}{A_V^2} = \frac{4kT\gamma}{g_{m3}} + \frac{4kT}{g_{m3}^2 R_D} + \frac{4kT\gamma g_{m1}}{g_{m3}^2}$$

$$= 4kT \left(\frac{\gamma}{g_{m3}} + \frac{1}{g_{m3}^2 R_D} + \frac{\gamma g_{m1}}{g_{m3}^2} \right)$$

7.4 Derive the **input-referred thermal and 1/f noise power density** of the circuits shown in Fig. 7.4 and compare the results. Assume that the circuits draw equal power supply currents.

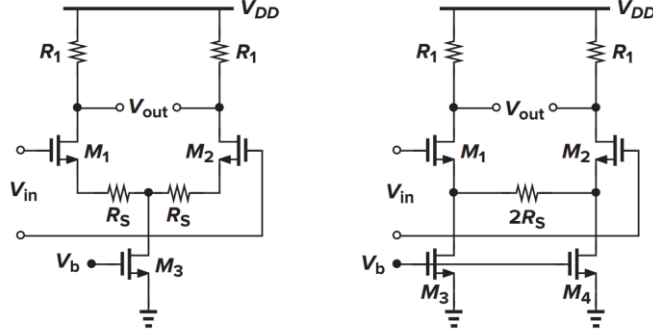


Fig. 7.4

第一幅图分析：

热噪声：

$$\overline{V_{n,in}^2} = 2 \left(\frac{4kT\gamma}{g_{m1}} + 4kTR_S + \frac{4kT(1 + g_{m1}R_S)^2}{g_{m1}^2 R_1} \right)$$

闪烁噪声：

$$\overline{V_{n,in}^2} = \frac{2K}{C_{OX}WL} \cdot \frac{1}{f}$$

第二幅图分析：

热噪声：

先对 M_3 和 R_S 产生的热噪声进行分析

在 M_3 源端产生的热噪声为：

$$\left(4kT\gamma g_{m3} + \frac{4kT}{R_S} \right) R_A^2, \text{ 其中 } R_A = R_S \parallel \frac{1}{g_{m1}} \parallel r_{o3} \approx R_S \parallel \frac{1}{g_{m1}} = \frac{R_S}{1 + g_{m1}R_S}$$

我们要先将这个噪声先等效到输出端，这个噪声到输出端的路径是一个共栅极放大器，故我们有：

$$\overline{V_{out,M3, R_S}^2} = g_{m1}^2 R_1^2 \left(4kT\gamma g_{m3} + \frac{4kT}{R_S} \right) R_A^2$$

$$\overline{V_{in,M3, R_S}^2} = \frac{g_{m1}^2 R_1^2 \left(4kT\gamma g_{m3} + \frac{4kT}{R_S} \right) R_A^2}{\left(\frac{g_{m1}R_1}{1 + g_{m1}R_S} \right)^2} = 4kT\gamma g_{m3} R_S^2 + 4kTR_S$$

剩下的热噪声和第一幅图相同，我们将单端噪声乘以 2，可以得到总的热噪声为：

$$\overline{V_{n,in}^2} = 2 \left(\frac{4kT\gamma}{g_{m1}} + 4kTR_S + \frac{4kT(1 + g_{m1}R_S)^2}{g_{m1}^2 R_1} + 4kT\gamma g_{m3} R_S^2 \right)$$

闪烁噪声： M_3 ， M_4 产生的闪烁噪声等效到输入端的方法与热噪声相同，这里不再赘述。

$$\overline{V_{n,in}^2} = \frac{2K}{C_{OX}W_1L_1} \frac{1}{f} + \frac{2K}{C_{OX}W_2L_2} \frac{1}{f} g_{m3}^2 R_S^2$$

7.5. Assuming that $\lambda = \gamma = 0$, derive the expression of the input-referred **thermal noise power density** for each circuit in Fig. 7.5 For part (a), assume that $g_{m3,4} = 0.5g_{m5,6}$.

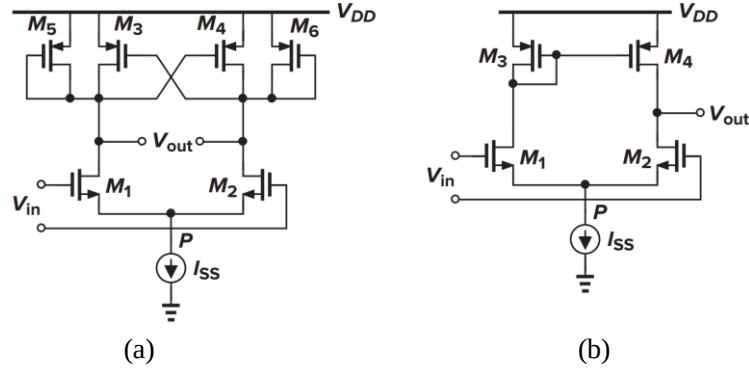


Fig. 7.5

(a) 先利用半边电路等效法，分析一侧噪声

$$\overline{V_{n,out1}^2} = 4kT\gamma(g_{m1} + g_{m3} + g_{m5})R_{out}^2 = 4kT\gamma(g_{m1} + 1.5g_{m5})R_{out}^2$$

$$\overline{V_{n,in1}^2} = \frac{\overline{V_{n,out1}^2}}{A_V^2} = \frac{4kT\gamma(g_{m1} + 1.5g_{m5})R_{out}^2}{g_{m1}^2 R_{out}^2} = \frac{4kT\gamma(g_{m1} + 1.5g_{m5})}{g_{m1}^2}$$

而后我们将输入单边输入噪声乘以 2 就可以了：

$$\overline{V_{n,in}^2} = 2\overline{V_{n,in1}^2} = \frac{4kT\gamma(2g_{m1} + 3g_{m5})}{g_{m1}^2}$$

(b) 没有出现差分时，五管单元是完全对称的，所以可以按照完全对称的差分电路进行分析

$$\overline{V_{n,in}^2} = \frac{4kT\gamma(2g_{m1} + 2g_{m4})}{g_{m1}^2} = 4kT\gamma\left(\frac{2}{g_{m1}} + \frac{2g_{m4}}{g_{m1}^2}\right)$$